

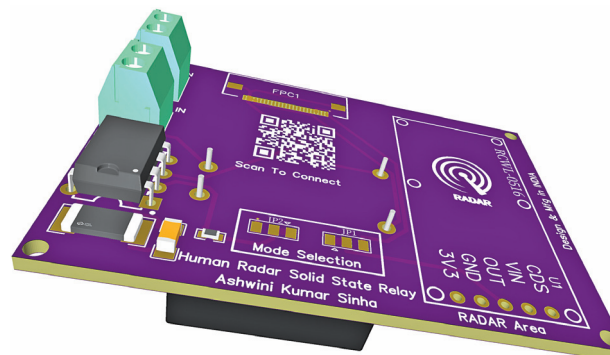
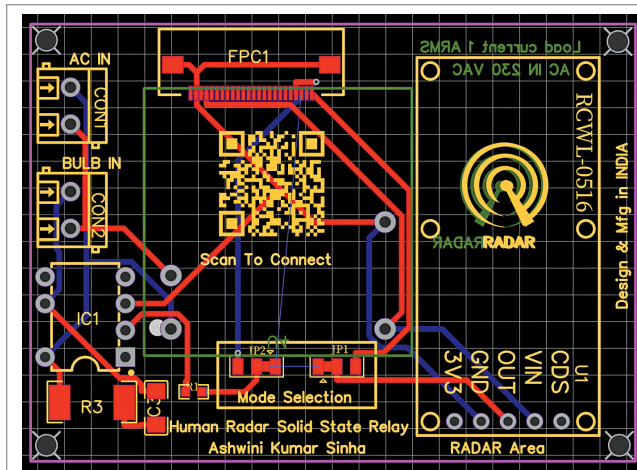


Fig. 8: Final PCB design

IndusBoard sends control signals wirelessly, for example, through an MQTT app or IoT dashboard, to switch the connected AC load (for example, a bulb) on or off. In this mode JP2 = 2 and 3 shorted.

A built-in snubber network ensures clean and stable AC switching. JP2 shorted across pins 1 and 2 enables remote operation of the SSR via IndusBoard signals, allowing IoT-based control of the connected load.

Mode 4: Human radar IoT SSR + sensor mode. In this hybrid mode, the circuit combines both automatic motion-based control and IoT-based monitoring for advanced automation. Human motion triggers local SSR switching while simultaneously sending status updates to the IndusBoard through the FPC connector, providing both sensor and light-state feedback. All jumpers shorted.

Configure JP1 shorted (radar OUT to optotriac input) and SJ1 bridged (OUT to IO3 for sensor feedback). JP2 may also be shorted if remote control

via IO4 is required. The radar's OUT signal is routed in two ways: locally to the optotriac for immediate switching, and to the IndusBoard's FPC pin IO3 for alerts such as 'motion detected' notifications. The light (opto) state can be mirrored back via IO4 for real-time IoT dashboard updates.

An optional LDR can be connected to the CDS pin for night-only activation.

Refer to Fig. 2 for the dual-path configuration (JP1 and SJ1 shorted) and Fig. 3 for FPC pin assignments (pins 3-4 active, VDD/GND on pins 25-26 for 3.3V/5V compatibility).

Mode 4 integrates radar sensing with IoT control. The radar output drives the local SSR and simultaneously communicates motion and light states to the IndusBoard for monitoring or remote control.

The four modes above represent different functional settings. Fig. 7 shows the final circuit of the motion sensor system. For powering the radar sensor, IndusBoard, and opto-isolator, a DC supply is derived through an AC-DC converter mounted on the PCB. The AC input is connected to the screw terminal CON1, while the load, such as a bulb or low-power fan, is connected to CON2.

PCB design

The final system's PCB design has been kept as compact as possible by

using a multilayer layout and SMT components. The top layer includes the radar module, optotriac, and other discrete components, while the bottom layer houses the AC-DC power converter and connectors.

The compact multilayer PCB layout separates AC and DC sections to ensure safety and minimise interference, with SMT components used for space efficiency.

Testing

For testing, first assemble all the components on the PCB as shown in Fig. 1. Connect the AC power line to the screw terminal CON1 and the load, such as a bulb, to CON2. Based on the solder jumper configuration, if set to a particular mode, the system automatically turns the light on or off according to motion detection. When an LDR is added to the radar sensor module, the circuit operates according to motion only during low-light or night conditions. The same principle applies to the other mode configurations.

For IoT functionality, connect the IndusBoard Coin using the FPC connector.

The IoT code and app development for this system will be covered in a subsequent article to be published soon. **EFY**

Ashwini Kumar Sinha, an IoT and AI enthusiast, is a Tech Journalist at EFY.

TO DOWNLOAD SOURCE CODE AND RESOURCES



Scan this QR code or go to <https://sourcecode.electronicsforu.com> to access and download source code and other resources for this project